Healthcare Clinic Database System

Maximiliano Armenta, Paul Whitby, David Barrios

California State University, Fullerton

CPSC 332: File Structures and Database Systems

Ken Wuttisasiwat

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System Description

The Healthcare clinic database system is a relational database designed to support the daily operations of a healthcare clinic. The system centralizes patient records, appointment scheduling, billing, and staff management into a single structured database. It replaces fragmented or paper-based processes with a digital solution that improves efficiency, accuracy, and accessibility of medical and administrative data.

The system will be used by clinic staff, including receptionists, doctors, and administrators. Receptionists will schedule appointments and manage patient records, doctors will access patient histories and update visit notes, and the administrators will oversee billing and reporting. The system tracks patients, appointments, medical services, payments, and staff assignments while ensuring proper relationships between these entities.

Scope

- The system will track patient demographic and contact information
- The system will schedule and manage appointments between patients and doctors
- The system will record medical services provided during each appointment
- The system will store doctor and staff information
- The system will process and track payments for services used
- The system will prevent scheduling conflicts for doctors
- The system will maintain historical records of patient visits
- The system will assign multiple services to a single appointment

Business Rules / Constraints

- Doctor and patient emails must be unique within their respective tables
- Patient Medical Record No./ID must be Unique
- EmployeeID and PatientID must be unique primary identifiers and in their respective tables
- Patient must have a first_name and last_name that are not NULL
- Phone numbers must be unique and not NULL
- The doctor's time slot must be one of [Available, Booked]
- No double-booking for the same doctor and time
- discharge_date must occur after admission_date
- PatientID in MEDICAL_RECORD must reference an existing PATIENT record
- Patient Appointment_status must be [Pending, Cancelled, Complete]

UML Overview (High-Level)

The healthcare clinic database is built around eight main tables: PATIENT, STAFF, SERVICE, APPOINTMENT, MEDICAL_RECORD, APPOINTMENT_SERVICE, INVOICE, and PAYMENT. These tables represent the core parts of the clinic's workflow, including patient information, staff assignments, medical services, appointment scheduling, visit records, billing, and payments.

The PATIENT table stores patient demographic and contact information, with example attributes like patient_id, first_name, last_name, email, and phone. The STAFF table stores employee information, with attributes like staff_id, first_name, last_name, email, phone, and role. Staff roles are limited to Doctor, Receptionist, and Administrator to match the main users described in the system. The SERVICE table stores medical services provided by the clinic, with example attributes including service_id, service_name, and service_cost.

The APPOINTMENT table stores scheduled visits between patients and doctors, with attributes such as appointment_id, appointment_date, appointment_time, and appointment_status. Appointment status is restricted to the values Pending, Cancelled, and Complete. The MEDICAL_RECORD table stores visit information, including record_id, diagnosis, visit_notes, admission_date, and discharge_date, which supports the system's goal of maintaining patient histories and visit notes. Billing information is handled through the INVOICE and PAYMENT tables. The INVOICE table stores billing records for appointments, while the PAYMENT table stores payments made toward those invoices.

The relationships between these entities are as follows:

PATIENT ➡ APPOINTMENT (1:N)

A PATIENT can have many APPOINTMENT records, while each APPOINTMENT belongs to one PATIENT (1:N).

STAFF ➡ APPOINTMENT (1:N)

A STAFF member can be assigned to many APPOINTMENT records, while each APPOINTMENT is assigned to one STAFF member (1:N).

PATIENT ➡ MEDICAL_RECORD (1:N)

A PATIENT can also have many MEDICAL_RECORD entries, while each MEDICAL_RECORD belongs to one PATIENT (1:N)

APPOINTMENT ➡ MEDICAL_RECORD (1:1)

Each APPOINTMENT is tied to one MEDICAL_RECORD (1:1)

APPOINTMENT ➡ INVOICE (1:1)

Each APPOINTMENT generates one INVOICE (1:1)

INVOICE ➡ PAYMENT (1:N)

An INVOICE can have many PAYMENT records, while each PAYMENT belongs to one INVOICE.

The system also includes a many-to-many relationship between APPOINTMENT and SERVICE. One appointment may involve more than one service, and one service may be provided during many appointments. This relationship is resolved through the junction table APPOINTMENT_SERVICE, which connects appointment IDs and serviceIDs. This satisfies the requirement to include at least one M:N relationship with a junction table.

Overall, the design includes eight tables, which exceeds the minimum requirement of six. Core operationable tables include PATIENT, STAFF, SERVICE, APPOINTMENT, MEDICAL_RECORD, INVOICE, and PAYMENT, while APPOINTMENT_SERVICE serves as the junction table that supports the many-to-many relationship between appointments and services.